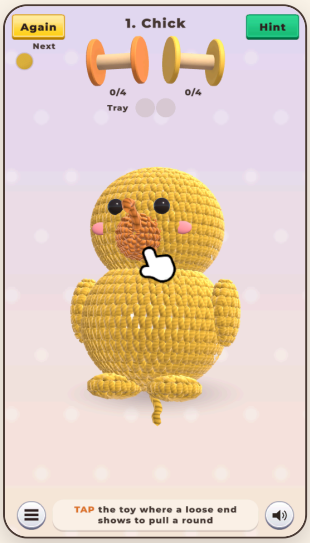
GAME DESIGN DOCUMENT · 0.2.0

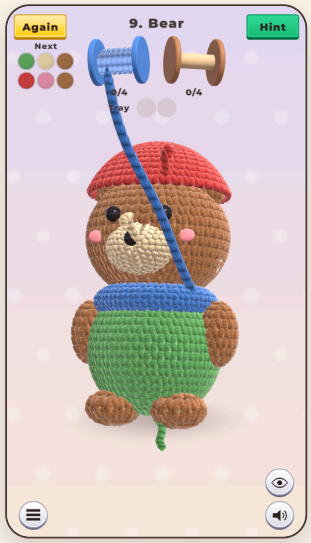
Unstitched

Crocheted toys come apart the way real ones do, round by round. Each round has to land on an open spool of its colour or wait in a tiny tray, so the order you pull in is the puzzle.

AT A GLANCE	
Genre	Hyper-casual 3D unravel puzzle; inspired by Yarn Fever! Unravel Puzzle
Platforms	Portrait phones and the browser (9:16); Unity 6, URP
Controls	Tap a part to pull a round; hold to keep pulling; drag to turn the toy
Session	A toy is 12 to 40 rounds; there is no timer
Content	13 toys, each adding one idea; all open from the level picker
Status	Playable prototype 0.2.0: web build



Level 1: the hand teaches a tap



Level 9: a round reels onto its spool



Level 13: rounds wait in the tray

The core loop

- **Look:** turn the toy; which parts are free, and what colours come next in the queue?
- **Pull:** a round comes off stitch by stitch and reels onto its spool.
- **Fill:** four rounds fill a spool; it pops and the next in the queue opens.
- **Unravel:** take every round off. If nothing can be pulled, the toy tangles.

Design pillars

- **It comes apart like crochet.** Real stitches and rounds, sewn-on parts, stuffing behind.
- **Every rule shows on the toy.** A loose end means free; a stuck part says what is sewn on it.
- **Hidden, but fair.** Layers hide colours; the eye shows one layer down.
- **Difficulty is measured.** Every position of every toy is visited, and the spool order is chosen from that.

Who it is for

Players of calm sorting and unravel puzzles who enjoy the texture of the thing itself. The market's yarn puzzles remove whole patches of a model; here the toy is crocheted, so a part has a loose end, rounds leave one stitch at a time and the stuffing shows behind them.

RULES

Parts, rounds, spools and the tray

The toy

A toy is made of **parts** (a hat, a head, a sweater), and each part is a stack of **rounds** in one or more colours. Rounds come off from the last one made. A part's next round shows as a curl of loose yarn.

Sewn on

A part can be **sewn on** another: the part underneath can lose only some of its rounds until the one on top has gone. The muzzle must leave before the head, the roofs before the towers. Tapping a stuck part names what is sewn on it, and that part swells.

Spools, the queue and the tray

Two **spools** are open at a time, each taking four rounds of its colour; the rest wait in a **queue** the player can read. A full spool leaves and the next opens. A round with no open spool of its colour waits in the **tray**, which has 0 to 3 places, and hops onto its spool when one opens. A pull that needs a place when the tray is full is refused, and the empty wells flash red.



A held part says what is sewn on it

Winning and tangling

Take every round off and the toy is **Unravelled**; the card says how many rounds waited in the tray. If no part can be pulled, it is **Tangled** and Again starts it over. A hint makes a part on a winning line swell, or says the toy can't be saved from here.

Layers and the eye

From level 5 a **layer** (a dress, a raincoat, a sweater, icing, wrapping, a bigger doll) covers other parts, and what is under it can't be seen. Holding the **eye** button turns layers see-through. It shows **one layer down**: on the doll, the middle doll stays solid until the big one is off, so the smallest is never given away.

> **Why one layer down.** Layers make a colour a surprise; peeking makes the surprise fair. Showing everything at once would turn the doll back into a toy with no layers, so the eye answers only the next question.



The eye: under the sweater

Two ways to pull

A **tap** takes one round. **Holding** keeps pulling, one round after another, until the finger lifts. A press stays undecided until it moves (a drag turns the toy), lasts 0.22 seconds (a hold) or lifts (a tap). The penguin, level 2, can only be won by stopping early: a pull that takes a whole strand can't win it.

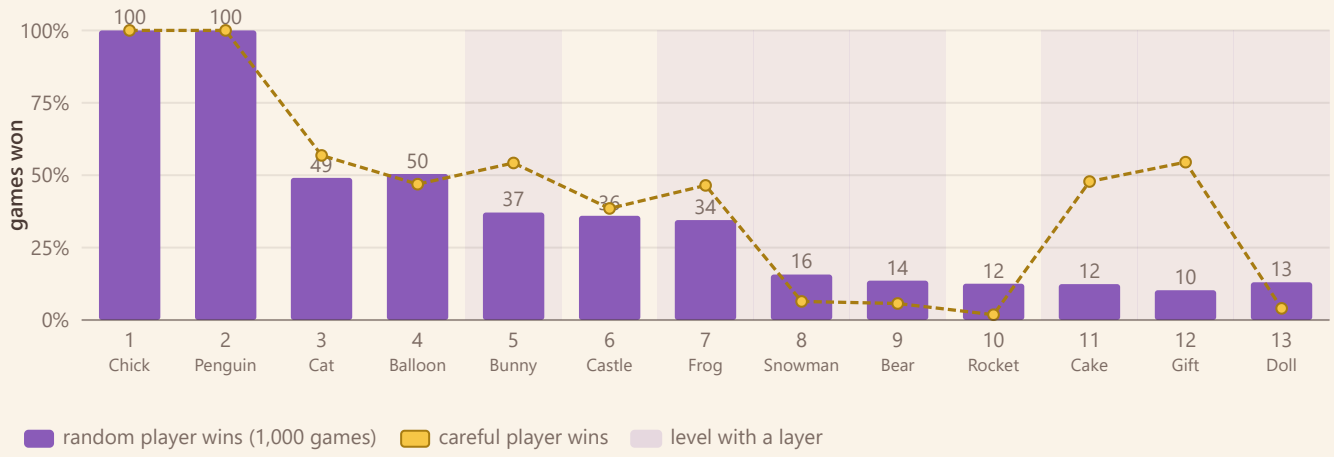
LEVEL DESIGN

Thirteen toys, one new idea each

Each toy brings in one idea, and the order is set by how forgiving each toy is. Layers start at level 5 with the eye; the castle has none, as a breather, and the rocket trades a layer for five loose ends at once.

TOY	NEW IDEA	PARTS	ROUNDS	COLOURS	TRAY	LAYERS	RANDOM	CAREFUL
1. Chick	Pulling; the beak comes off the head first	3	12	2	2	–	100%	100%
2. Penguin	Stopping early: no tray, so take only what the spool holds	3	16	3	0	–	100%	100%
3. Cat	The tray, and a chain of sewing: bell, collar, body	6	24	3	2	–	49%	57%
4. Balloon	An object; it comes apart from the bottom up	4	28	5	1	–	50%	47%
5. Bunny	The first layer: a dress over the stripes; the eye	6	32	4	3	1	37%	54%
6. Castle	No layer, a breather: roofs before towers	8	28	5	2	–	36%	38%
7. Frog	One place in the tray, and a raincoat	7	32	5	1	1	34%	46%
8. Snowman	A coat with a colour under it you won't guess	6	32	6	2	1	16%	6%
9. Bear	A sweater; pulling whatever fits loses	8	32	6	2	1	14%	6%
10. Rocket	No layer, five loose ends at once	6	24	4	2	–	12%	2%
11. Cake	Two layers of icing over two sponges	8	36	7	3	2	12%	48%
12. Gift	A teddy inside the wrapping	7	40	6	3	1	10%	55%
13. Doll	A doll in a doll in a doll; the eye shows only the next	7	40	5	2	2	13%	4%

The difficulty curve: how often a player who doesn't think wins



The **random player** pulls any legal round; the **careful player** pulls a colour with an open spool whenever one shows and sees only what is showing. Neither is a person: they measure how forgiving a toy is and how fair it is without peeking. Where the careful player does far worse than random (snowman, bear, rocket, doll), thinking ahead matters most.

LEVEL DESIGN

How difficulty is measured and chosen

1 · The toy: what I decide

A toy is data: its parts, the colours of their rounds, what is sewn on what and how many rounds stay free, and which parts are layers. The idea it adds decides the shape; a castle comes down roofs first because roofs sit on towers.

2 · Every position, visited

`ToyReport` visits every position a toy can reach and walks back from the wins, so it knows which positions are already lost. From that come the random player's and the careful player's win rates, the share of positions lost, and how many first pulls lose. A solver finds the line that sends the fewest rounds to the tray.

3 · The tuner chooses the queue

The same parts with a different spool order, or a different tray, is a different level. The tuner measures orders and tray sizes and writes them out; I pick the one that fits the toy's place on the curve. Toys with long chains of sewing have so few winnable orders that a random sample missed them all, so the tuner finds them by swapping two spools at a time.

4 · What the numbers can't say

Both players are bots. The random player says how forgiving a toy is, the careful one how fair it is to someone who hasn't peeked; neither reads ahead the way a person does. They set the order until first-time players can be watched.

Finding: most orders are too safe

For the shipped tray, every order (cat) or a sample of 80 winnable orders found by swapping spools from the chosen one. Most of them can't be lost at all, and nearly all are more forgiving than the one in the game: the queue has to be chosen on purpose.

TOY	TRAY	ORDERS	MEASURED	UNLOSABLE	EASIER	CHOSEN
3. Cat	2	30	25	88%	22	53%
5. Bunny	3	840	80	55%	69	40%
7. Frog	1	3,360	80	80%	79	31%
9. Bear	2	6,720	80	84%	78	23%
11. Cake	3	90,720	80	61%	79	15%
13. Doll	2	75,600	80	65%	78	11%

Orders: every different order of the toy's spools. **Measured:** winnable orders measured. **Unlosable:** their share with no lost position. **Easier:** how many a random player wins more often than the chosen one. **Chosen:** the game's order, random wins over 100 games as the tuner measures. The workbook has all 13 toys.

Tests that guard the design

- Every toy is well made (each round has a spool place, sewing never goes in a circle) and can be won.
- The chick can't be lost, the penguin needs stopping early, and each toy from the balloon on is about as hard as the last, or harder.
- A PlayMode test wins every toy in the real scene with the solver's line and saves screenshots; others check the eye, the tray warning and the doll's layers.

The thirteen toys, in play order



ITERATIONS

What the measurements and the builds showed

Unstitched has not been playtested by first-time players yet. These changes came from the tuner, from playing each build and from comparing it with the game that inspired it.

1 · The first test toy held no decision

The concept's toy taught stopping early but could not be lost once you stopped. Over every order of the bear's spools with trays of 1 to 4, 462 setups could be won and 327 of them could not be lost at all.

> **Change:** difficulty became a measured choice. Each toy's spool order and tray come from the tuner, and a test pins the shape of the curve.

2 · Tags or the toy itself

Pulls first went through tail tags on a ring around the toy. Yarn Fever lets you tap the toy itself, so both were built with a switch and compared.

> **Change:** I chose tapping the toy; the tags went, a curl of loose yarn now marks each free part, and a press waits to see if it is a tap, a hold or a drag.

3 · Layers, tried on one toy first

A sweater went on the bear alone, to see whether hidden layers were fun before reworking every level. Peeking at everything would undo the layer, and a tray that glowed from the start of a one-place level read as an alarm.

> **Change:** layers run from level 5 to the end, the eye shows one layer down, and the tray warns only once something is waiting in it.

4 · The gift and the doll were re-sewn

The first gift and doll could hardly be lost whatever the order. They were re-sewn with more loose ends at once and shared colours, which gave the tuner orders worth choosing.

5 · A bigger sample put the rocket out of order

The tuner chose with 100 random games an order. Measured again with 1,000, the curve still fell, but the rocket (20%) came out clearly easier than the bear before it (14%), beyond both margins of error. 100 games rank one toy's orders well enough; placing two close toys needs the bigger sample.

> **Change:** every rocket order with trays of 1 to 3 was measured at 1,000 games. Swapping its last two spools (blue before red, same tray) gives 12.5%, between the bear (13.6%) and the cake (12.4%), and the careful player falls to 1.8%.

6 · The web build drew nothing

Found by running it, not by the tests: the browser rejected every draw call over the shadow map, and Unity's sphere needs the physics module that web builds strip. Shadows are off and spheres are made in code; stuffing was rebuilt twice after looking at it, since lumps read as snowballs and a fibre photo as a smooth egg.

FEEL, LOOK AND NEXT STEPS

Juice, art, monetisation and the roadmap

Game feel

- A round's stitches leave one at a time from the side facing the camera, about 62 a second; the next nine lift before they let go.
- A thick 3D rope shoots out on a pull, ripples, and its tail whips into the spool.
- Spools spin as yarn winds on, squash on landing and pop when full; the next springs in.
- The toy rocks on each pull and more on a refusal; a win rains yarn balls.
- The interface follows the yarn: a spool or the tray changes only when the yarn arrives.

Art and audio

No 3D models: every toy is built from its stitches in code, each stitch a V of two loops (1,058 stitches on the chick, 2,971 on the cake). A hand-written wool shader uses fibres from Poly Haven's Caban texture; stuffing is billowing noise over eight shells. Music and jingles by MintoDog, buttons and click from Kenney, type in Montserrat.

Built with

Unity 6 (URP). The rules, the solver and the report live in a pure C# library with no engine code, with EditMode and PlayMode tests. Built with an AI coding assistant (Claude): I set the rules, the toys, their order and the tuning, played every build and decided each change.

Monetisation (a plan, not built)

The prototype has no ads or purchases. As a hyper-casual release it would follow the genre:

- **Interstitials** after a toy, never mid-toy, and not in the first four.
- **Rewarded extra tray place** when a toy tangles, to carry on instead of starting over.
- **Remove ads** as a single purchase; new toys in packs later.

What to measure first

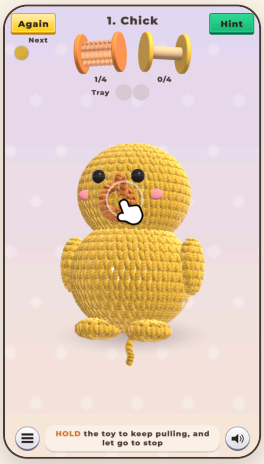
- Day-1 retention and the toy where players stop: does the curve hold?
- Tangles and hints per toy: which toys are walls?
- Eye use on the bunny (level 5): do players find it unaided?

Roadmap

- Playtests with first-time players; set the order from what they show, not from the bots alone.
- Measure on a real phone: the frame rate has been checked only on a laptop (61 fps in a browser on a resting toy).
- An Android build, then more toys and themes from the same parts.

Credits

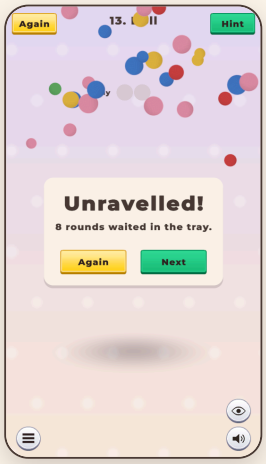
Music and jingles: MintoDog, “Cozy Puzzle” (OpenGameArt.org, CC0). Wool fibres: “Caban”, Poly Haven (CC0). Buttons, click and pointer: Kenney (kenney.nl, CC0). Font: Montserrat (SIL Open Font License).



The hand teaches a hold



A full spool, about to pop



Unravelled: rounds that waited



Every toy open from the picker